

4th Subject

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Structured Query Language

Consider the following schema:

Suppliers(sid: integer, sname: string, address: string)

Parts(pid: integer, pname: string, color: string)

Catalog(sid: integer, pid: integer, cost: real)

The key fields are underlined, and the domain of each field is listed after the field name. Therefore sid is the key for Suppliers, pid is the key for Parts, and sid and pid together form the key for Catalog. The Catalog relation lists the prices charged for parts by Suppliers. Write the following queries in Structured Query Language SQL

1. Find the names of suppliers who supply some red part.
2. Find the sids of suppliers who supply some red or green part.
3. Find the sids of suppliers who supply some blue part or are at Cluj-Napoca.
4. Find the sids of suppliers who supply some black part and some white part.
5. Find the sids of suppliers who supply every part.
6. Find the sids of suppliers who supply every black part.
7. Find the sids of suppliers who supply every black or white part.
8. Find the sids of suppliers who supply every black part or supply every white part.
9. Find pairs of sids such that the supplier with the first sid charges more for some part than the supplier with the second sid.
10. Find the pids of parts supplied by at least two different suppliers.
11. Find the pids of the most expensive parts supplied by suppliers named Vasile.
12. Find the pids of parts supplied by every supplier at less than 500 lei. (If any supplier either does not supply the part or charges more than 500 lei for it, the part is not selected.)